Lesson 6: KINGDOM ANIMALIA

PHYLUM PORIFERA

Phylum Porifera, commonly known as **sponges**, represents the simplest group of multicellular animals. They are evolutionary significant because they represent the transition from colonial unicellular organisms to true multicellular life.

General Characteristics of Poriferans

- Include a system of pores (also called Ostia) and canals through which water passes.
- Sponges are distinct from all other animals in that they lack true tissues, organs, and complex body symmetry.
- **Cellular Level of Organization:** Instead of tissues, they have specialized cells that perform specific functions.
- **Asymmetry:** Most sponges lack any formal symmetry (though some have radial symmetry).
- **Sessile:** Adult sponges are permanently attached to a substrate (rocks, coral reefs, etc.) and do not move.
- **Filter Feeders:** They are "living pumps" that draw water through their porous bodies to trap microscopic food particles.
- Sponges do not have nervous, digestive or circulatory systems instead most rely on maintaining a constant water flow through their bodies to obtain food and O2and remove wastes.
- Most species reproduction by sexual reproduction and other by asexual reproduction (***Fragmentation:*** *If a piece of the sponge breaks off, it can reattach to a new surface and grow into a complete, genetically identical individual. This is a testament to the high degree of cellular independence in sponges.* ***Budding:*** *A small "bud" grows out of the parent, eventually detaching to become a new sponge). Freshwater sponges as well as several marine species from resistant structures called gemmules that can with stand adverse, conditions such as drying or cold and later develop into new individuals. Gemmules: Are aggregates of sponges tissue and food covered by a hard coating containing spicules or sponging fibers.).*

Sexual Reproduction

Sponges are generally **hermaphroditic**, meaning a single individual can produce both eggs and sperm, though they usually don't self-fertilize.

Sperm Release: Sponges are "broadcast spawners." They release massive clouds of sperm into the water column through their osculum.

Internal Fertilization: When sperm from one sponge enters another sponge through its pores (ostia), it is captured by the **choanocytes**. The choanocyte loses its collar and flagellum, transforms into an amoebocyte, and carries the sperm to the egg located in the middle layer (mesohyl).

Larval Stage: The fertilized egg develops into a tiny, flagellated larva that is released into the water. Unlike the adult, the larva is free-swimming. It swims until it finds a suitable surface, attaches itself, and undergoes a dramatic metamorphosis to become a sessile adult sponge.

The Unique Body Plan

The sponge body is designed around a water-flow system.

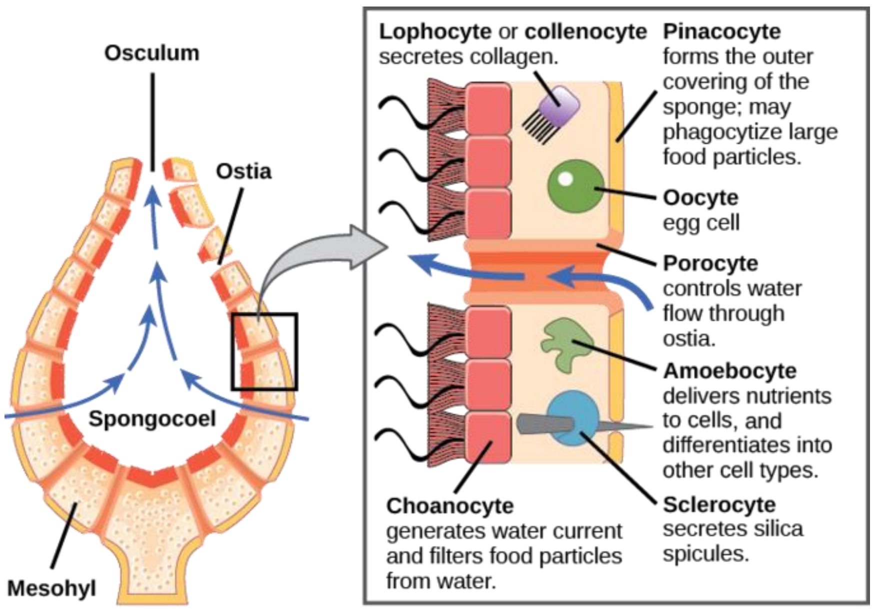
Ostia: Tiny pores on the outer surface where water enters the sponge.

Spongocoel: The large, central cavity of the sponge.

Osculum: The large opening at the top where filtered water exits.

Choanocytes (Collar Cells): These are the most critical cells. They have a flagellum that beats to create a water current and a "collar" that traps food particles from the passing water.

Amoebocytes: These cells roam through the sponge's middle layer (mesohyl), digesting food, distributing nutrients, and producing skeletal structures.



(a) Basic sponge body plan (b) Some sponge cell types

Skeletal Support

To maintain their shape, sponges produce a skeleton made of materials that also serve as a defense mechanism:

Spicules: Needle-like structures made of **calcium carbonate** (chalk-like) or **silica** (glass-like).

Spongin: A flexible, fibrous protein that provides structural support (this is the material found in natural bath sponges).

Importance of Sponges

Ecological Niche: They serve as habitats for many other marine organisms and are efficient at recycling nutrients in reef ecosystems.

Biochemical Potential: Because they are sessile and cannot flee, sponges produce a wide array of chemicals for defense. Many of these compounds are currently being researched for their **anti-cancer, anti-inflammatory, and antibiotic** properties.

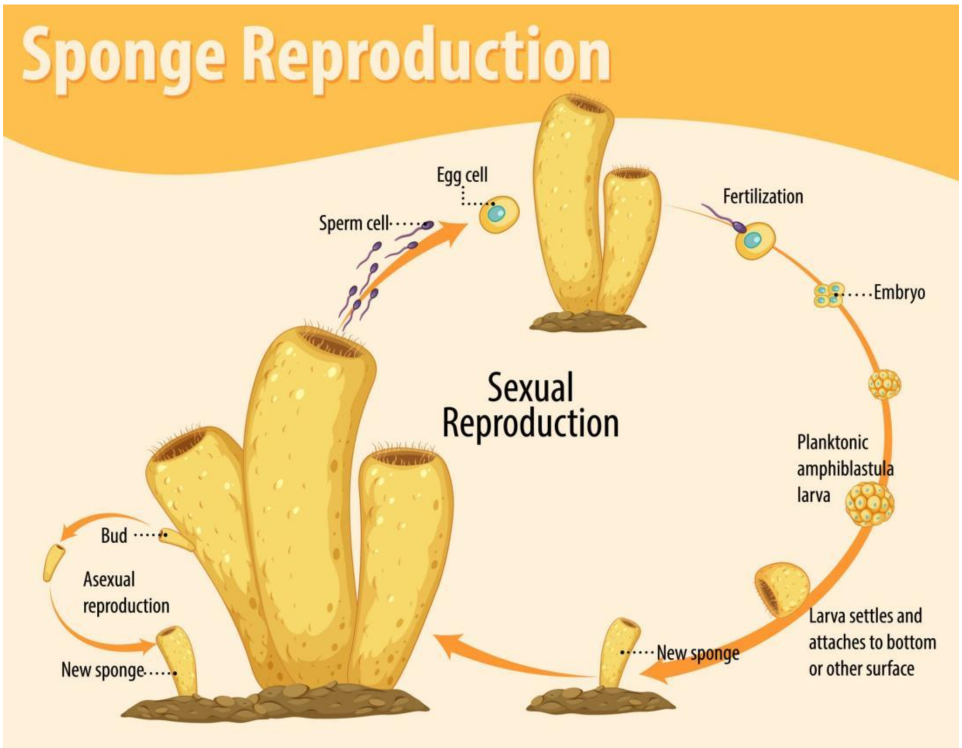


Diagram showing sponge reproduction

PHYLUM CNIDARIA

Phylum **Cnidaria** (pronounced *ny-DARE-ee-uh*) is a diverse group of aquatic animals that includes jellyfish, sea anemones, corals, and hydras. They are evolutionarily more complex than sponges because they possess **true tissues**, a nervous system, and a defined body cavity for digestion.

Whereas the defining cell type for the sponges is the choanocyte, the defining cell type for the cnidarians is the **cnidocyte**, or stinging cell. These cells are located around the mouth and on the tentacles, and serve to capture prey or repel predators. Cnidocytes have large stinging organelles called **nematocysts**, which usually contain barbs at the base of a long coiled thread.

<u>**General Characteristics of Cnidarians**</u>

- **Radial Symmetry:** Their body parts are arranged around a central axis, like a wheel. This allows them to detect stimuli (food or danger) from any direction.
- **Cnidocytes (The Defining Feature):** This is where their name comes from. They possess specialized stinging cells called **cnidocytes**, which contain a coiled, spring-loaded structure called a **nematocyst**. When triggered by touch, the nematocyst fires a harpoon-like thread to capture prey or defend against predators.
- Are diploblastic, meaning that they develop from two embryonic layers, ectoderm and endoderm. Nearly all (about 99 percent) cnidarians are marine species.

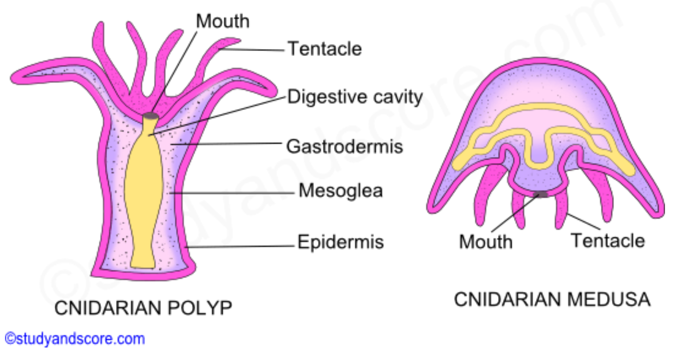
Gastrovascular Cavity: A single opening serves as both the mouth and the anus. They digest food inside this central sac.

2. Two Body Forms: The Dimorphic Life Cycle

Most cnidarians exhibit two distinct body forms during their life cycle. A single species may alternate between these two stages:

Polyp: A sessile (stationary) form where the tentacles and mouth face upward (e.g., coral and sea anemones).

Medusa: A free-swimming form where the bell-shaped body and tentacles face downward (e.g., jellyfish).



3. Classification (Main Classes)

Class Anthozoa (Flowers of the sea): Includes corals and sea anemones. They exist only as **polyps** and are often reef-builders.

Class Scyphozoa (True jellyfish): The **medusa** is the dominant form in their life cycle.

Class Hydrozoa: Often alternate between polyp and medusa stages. Many form colonies, such as the *Portuguese Man o' War*.

Ecological Importance

Reef Formation: Stony corals are the foundation of reef ecosystems, providing habitat for roughly 25% of all marine life.

Predation: As efficient carnivores, they play a critical role in controlling plankton and small fish populations.

Symbiosis: Many corals have a vital symbiotic relationship with algae (zooxanthellae), which provide them with nutrients through photosynthesis in exchange for protection.